



Faculty of Engineering and Technology
Electrical and Computer Engineering Department
ENCS4130 – Computer Network Laboratory

Experiment No. 05

Dynamic Routing 3 – Path Vector Routing Protocols Border Gateway Protocol (BGP)

■ Objectives

- (a) Learn how to configure and verify IP routing on Cisco routers.
- (b) Understand the concepts of interior and exterior gateway protocols.
- (c) Explore autonomous systems and their role in networking.
- (d) Gain hands-on experience with dynamic routing using BGP.

■ Requirements

- (a) Cisco Packet Tracer software.
- (b) Four routers and one switch.
- (c) Six PCs and two network sniffers.

■ Pre-Lab Task

Students are required to build the network topology shown in Figure 4 and configure the IP addresses following the guidelines in Section 5.3.1. In addition, they must complete the OSPF and EIGRP routing protocol configurations described in Sections 5.3.2 and 5.3.3, respectively.

5.1 Introduction

Dynamic routing enables routers to automatically exchange routing information and update their routing tables without manual configuration. Dynamic routing protocols are classified into two main types: *Interior Gateway Protocols (IGPs)* and *Exterior Gateway Protocols (EGPs)*.

IGPs operate within a single **Autonomous System (AS)**, a collection of networks under the control of one organization. Protocols such as *Enhanced Interior Gateway Routing Protocol (EIGRP)*, *Open Shortest Path First (OSPF)*, and *Routing Information Protocol (RIP)* determine the best path based on specific metrics like hop count or bandwidth. Their primary goal is to provide fast convergence and efficient routing within an AS.

On the other hand, *EGPs* facilitate communication between different ASes. The most widely used *EGP* is the *Border Gateway Protocol (BGP)*. Unlike *IGPs*, which prioritize internal efficiency, *BGP* is designed for scalability, stability, and policy-based routing, making it essential for Internet-wide routing. Note that the Internet is composed of multiple interconnected ASes, each managed by different organizations. Figure 1 illustrates the classification of *IGPs* and *EGPs*. Previous experiments covered *IGPs*, while this experiment focuses on *EGPs*, specifically the *BGP*.

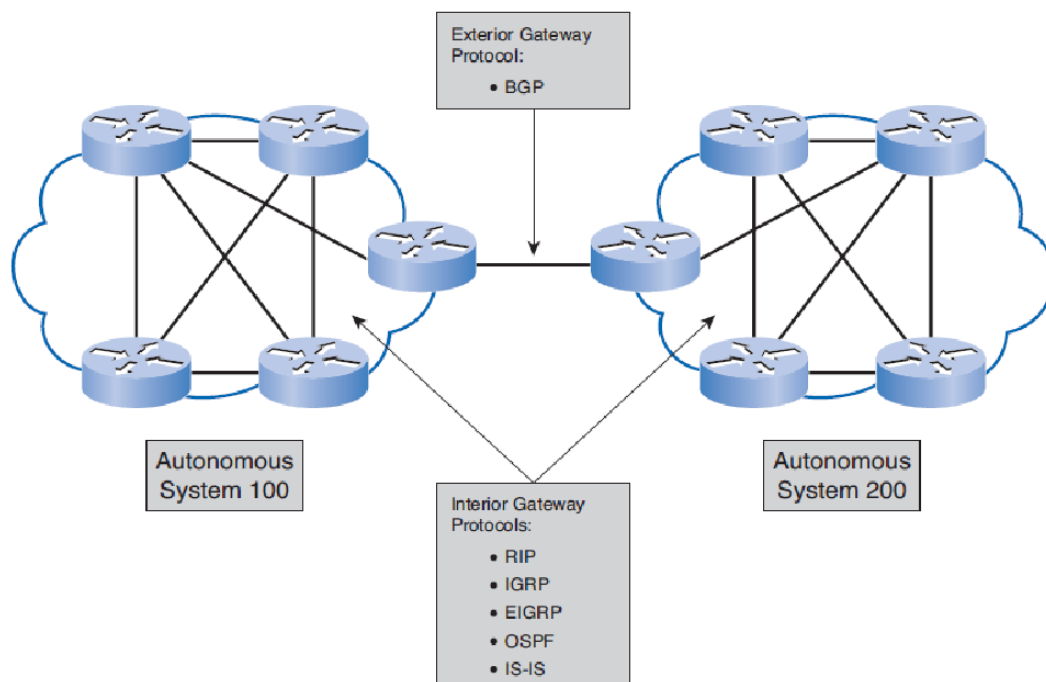


Figure 1: AS connectivity using BGP links.

5.2 Border Gateway Protocol (BGP)

BGP is a *Path Vector Protocol* designed for inter-domain routing. It makes routing decisions based on a set of attributes. The most critical of these is the **AS Path**, which records the sequence of ASes a route has traversed. Each AS is assigned a globally unique 16-bit AS number by the Internet Assigned Numbers Authority (IANA).

- Public AS numbers (1 – 64,511): Used for globally unique AS assignments.

- Private AS numbers (**64,512 – 65,535**): Reserved for internal use within organizations.

The path-based approach of BGP ensures loop prevention and enables flexible policy-driven routing decisions on the Internet:

- **Prevention of Routing Loops:** When a router receives a routing update, it checks the AS Path for its own AS number. If found, the route is discarded, preventing loops in which data would circulate endlessly between ASes.
- **Route Selection:** BGP selects the optimal path based on the AS Path and other attributes. Since AS Path information is exchanged between peers, routers can make informed decisions on the best route to a destination AS.

5.2.1 Using and Enabling BGP

BGP can be used to **manage outbound traffic**, **fault tolerance** (ensuring network availability by automatically rerouting traffic if a connection fails, preventing a single point of failure), and **redundancy** (having multiple paths for network traffic to ensure continued connectivity) when connecting to the Internet. However, when only a single connection to an external AS is used, BGP introduces unnecessary complexity without providing significant benefits. With over 100,000 routes on the Internet, it is important to avoid overburdening interior routers with excessive routing information. In many cases, an IGP like OSPF or EIGRP can efficiently handle outbound traffic.

The true advantage of BGP lies in ***controlling how traffic enters the local AS***. By influencing inbound routing policies, organizations can optimize performance, distribute traffic more efficiently, and enforce preferred routing paths from external networks. BGP becomes essential in the following scenarios:

- Multiple connections exist to external ASes (e.g., the Internet) through different providers.
- Multiple connections exist to external ASes through the same provider but with distinct routing policies or separate connection points.
- The routing equipment can handle the additional processing and memory demands of BGP.

To configure a BGP connection (based on the AS number to which the router belongs), use the following command:

```
Router(config)# router bgp <AS-NUMBER>
Router(config-router)#
```

5.2.2 BGP Communication

For BGP to function effectively, routers (also called BGP speakers) must establish stable and reliable *peer/neighbor relationships* to exchange network reachability information. When a BGP session is first established, peers exchange *full routing tables*. Afterward, only *incremental updates*—reflecting changes in network reachability—are sent, reducing unnecessary traffic. This contrasts with IGPs like RIP and OSPF, which frequently send

updates to maintain network topology. BGP's approach improves efficiency, particularly in large-scale networks like the Internet.

BGP relies on structured communication between peers to ensure accurate routing tables. Specifically, it uses specific messages and mechanisms to establish and maintain peer relationships, detect failures, and update routing information as needed. Key elements of BGP communication include neighbor establishment, message types, and the BGP Finite-State Machine (FSM), which governs session management.

A) BGP Peers/Neighbors

BGP peer relationships allow routers to exchange critical routing information and maintain accurate routing tables. There are two types of BGP neighbor relationships:

- **iBGP Peers:** These are BGP neighbors within the same AS. iBGP is used to exchange routing information between routers that belong to the same AS. iBGP routes have an *Administrative Distance (AD)* of 200, making them less trustworthy compared to EIGRP, OSPF, and RIP routes.
- **eBGP Peers:** These are BGP neighbors between different ASes. eBGP facilitates routing information exchange across ASes, typically over the Internet. eBGP routes have an AD of 20, making them more trustworthy than internal routes.

Note:

It is important to distinguish between an *IGP*, such as *OSPF* or *RIP*, and *iBGP*. Although both are used for routing, iBGP is a part of BGP specifically designed to distribute BGP-learned routes within a single AS, rather than calculating optimal paths like IGPs.

Figure 2 illustrates AS connectivity with BGP links. Routers B, C, and D, located in *AS 200*, can form an *iBGP* peer relationship among themselves. Meanwhile, Router A in *AS 100* and Router B in *AS 200* establish an *eBGP* peering.

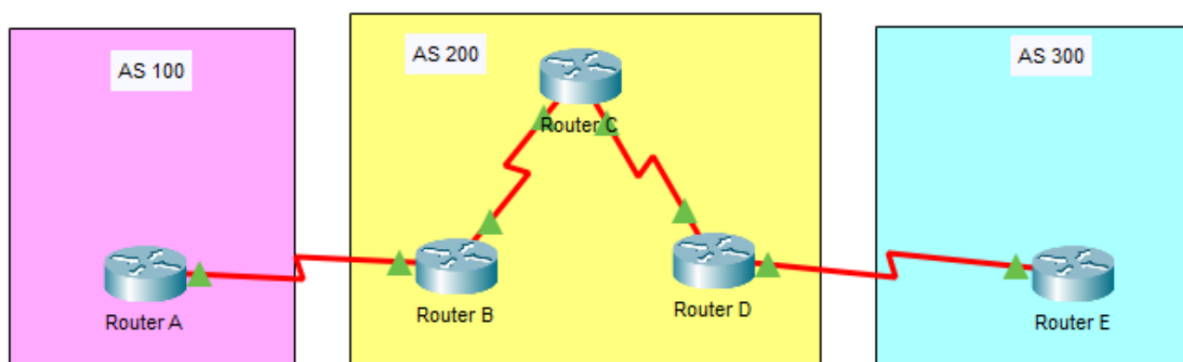


Figure 2: Example of AS connectivity with BGP links.

A Cisco router running BGP can belong to *only one* AS at a time. *IOS* enforces this restriction by allowing only one BGP process per router. To form a ***neighbor relationship*** between routers, use the following command:

```
Router(config-router)# neighbor <IP-ADDRESS-NEXT-INTERFACE>  
                           remote-as <AS-OF-REMOTE-NEIGHBOR>
```

Where:

- *IP-ADDRESS-NEXT-INTERFACE* refers to the IP address of the remote peer's interface.
- *AS-OF-REMOTE-NEIGHBOR* is the peer's AS number.

B) BGP Peers Messages

BGP establishes peer relationships and facilitates communication through a series of specialized messages, each serving a distinct purpose. These messages include:

- **OPEN Message:** This message is exchanged between peers to initiate a BGP session. It contains key parameters required for session establishment:
 1. *BGP Version:* Must match between peers for compatibility.
 2. *Local AS Number:* Specifies the AS number of the originating router.
 3. *BGP Router ID:* A unique identifier assigned to the router.
- **KEEPALIVE Message:** This message is sent periodically (default: every 60 seconds) to confirm that the peer is still active. If a router does not receive a *KEEPALIVE* message within the *Hold-Time* (default: 180 seconds), the remote peer is considered unreachable. To adjust the BGP Keepalive and Hold-time timers (in seconds) globally for all neighbors, use:

```
Router(config-router)# timers bgp <KEEPALIVE> <HOLD-TIME>
```

- **UPDATE Message:** This message is used to exchange routing information between peers. It contains details on network reachability and updates to the routing table.
- **NOTIFICATION Message:** This message is sent when a fatal error occurs in the BGP session. Upon receiving a NOTIFICATION message, the BGP session is immediately terminated, requiring re-establishment.

C) BGP Finite-State Machine (FSM)

BGP establishes a TCP session (port 179) with its peers and utilizes a FSM, as illustrated in Figure 3, to monitor and manage the operational status of each peer. The session progresses through multiple states, each defining a specific stage in the connection process:

- **Idle:** This is the initial state of the BGP session. BGP *waits* for a start event (e.g., manual configuration or a policy trigger). When a start event occurs, BGP moves to **Connect** state.
- **Connect:** In this state, BGP actively *initiates a TCP connection* to the peer. If the three-way TCP handshake *completes successfully*: (i) BGP resets the ConnectRetryTimer, (ii) Sends an *OPEN* message to the peer, and (iii) Moves to the

OpenSent state. If the handshake *fails* (e.g., no response from the peer) or the `ConnectRetryTimer` *expires*, BGP transitions to **Active** state.

- **Active:** BGP *waits for an incoming TCP connection request* from the peer. If a connection request arrives: (i) BGP establishes the session, (ii) Sends an *OPEN* message, and (iii) Moves to *OpenSent* state. If this attempt for TCP connection fails: (i) BGP resets the `ConnectRetryTimer`, and (ii) Moves back to *Connect* state.
- **OpenSent:** BGP has sent an *OPEN* message and is *waiting for the peer's OPEN message*. Once received, the router validates the message parameters (e.g., BGP version, AS number, etc.). If valid, a *KEEPALIVE* is sent, and the state moves to *OpenConfirm*. If errors exist, a *NOTIFICATION* is sent, and BGP resets to *Idle*.
- **OpenConfirm:** BGP waits for a *KEEPALIVE* or *NOTIFICATION* message. If a *KEEPALIVE* is received, the session transitions to *Established* state. If a *NOTIFICATION* is received or the *Hold Timer* expires, BGP resets to *Idle* state.
- **Established:** The session is fully functional, and peers exchange *UPDATE* messages to share routing information.

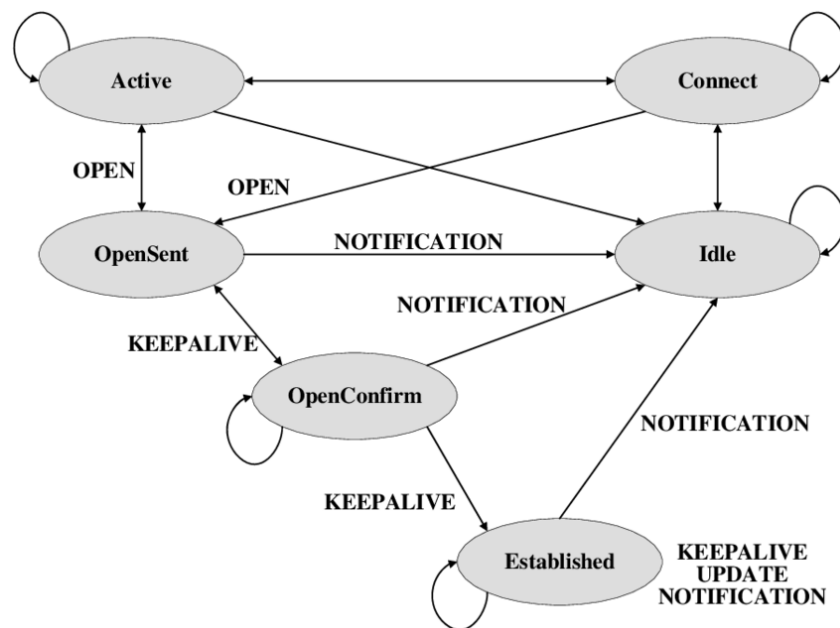


Figure 3: BGP finite-state machine and message flow.

Note that if a BGP session remains in the Active state, it may be due to one of the following reasons:

- Lack of IP connectivity between peers.
- Misconfigured neighbor statements (e.g., incorrect IP address or AS number).
- Firewall or Access Control List (ACL) rules block TCP port 179, preventing session establishment.

5.2.3 Route Redistribution Between BGP and OSPF/EIGRP

In many networks, BGP is used for external routing between ASes, while IGPs such as OSPF or EIGRP are used for internal routing within an AS. To allow these protocols to exchange routing information, redistribution must be configured.

A) Redistributing BGP into OSPF

To allow OSPF to learn routes from BGP:

```
Router(config)# router ospf <PROCESS-ID>
Router(config-router)# redistribute bgp <AS-NUMBER> subnets
```

- *PROCESS-ID*: OSPF process ID on the router.
- *AS-NUMBER*: BGP AS number running on the same router.

Note:

The “**subnets**” keyword is required. Without it, OSPF will only redistribute classful networks and ignore subnetted routes (e.g., /30).

B) Redistributing OSPF into BGP

To allow BGP to learn routes from OSPF:

```
Router(config)# router bgp <AS-NUMBER>
Router(config-router)# redistribute ospf <PROCESS-ID>
```

C) Redistributing BGP into EIGRP

To allow EIGRP to learn routes from BGP:

```
Router(config)# router eigrp <EIGRP-AS-NUMBER>
Router(config-router)# redistribute bgp <AS-NUMBER> metric
                        <BW> <DELAY> <RELIABILITY> <LOAD> <MTU>
```

- *EIGRP-AS-NUMBER*: EIGRP AS number on the router.
- *BW*: Minimum bandwidth along the path (in Kbps).
- *DELAY*: Total delay along the path (in tens of microseconds).
- *RELIABILITY*: Path reliability (0–255; 255 = fully reliable).
- *LOAD*: Path load/utilization (0–255; 1 = nearly idle).
- *MTU*: Maximum Transmission Unit of the path (in bytes).

D) Redistributing EIGRP into BGP

To allow BGP to learn routes from EIGRP:

```
Router(config)# router bgp <AS-NUMBER>
Router(config-router)# redistribute eigrp <EIGRP-AS-NUMBER>
```

5.3 Procedure

In this lab, we will connect four routers and multiple PCs across different networks. To enable communication between these networks, we will configure routing protocols on the routers. Specifically, we will implement OSPF and EIGRP for dynamic routing within the same AS and BGP for routing between different ASes.

5.3.1 Building the Topology

Construct the network topology shown in Figure 4, which consists of the following devices:

- Routers: Use **Router-PT**.
- Switches: Use **Switch-PT**.
- PCs: Use **PC-PT**.
- Packet Capture: Place a **Sniffer** between Router1 and Router2, and another between Router1 and PC2 to monitor incoming packets effectively.
- Connections: Use the “**Automatically Choose Connection Type**” option to link PCs, switches, and routers.

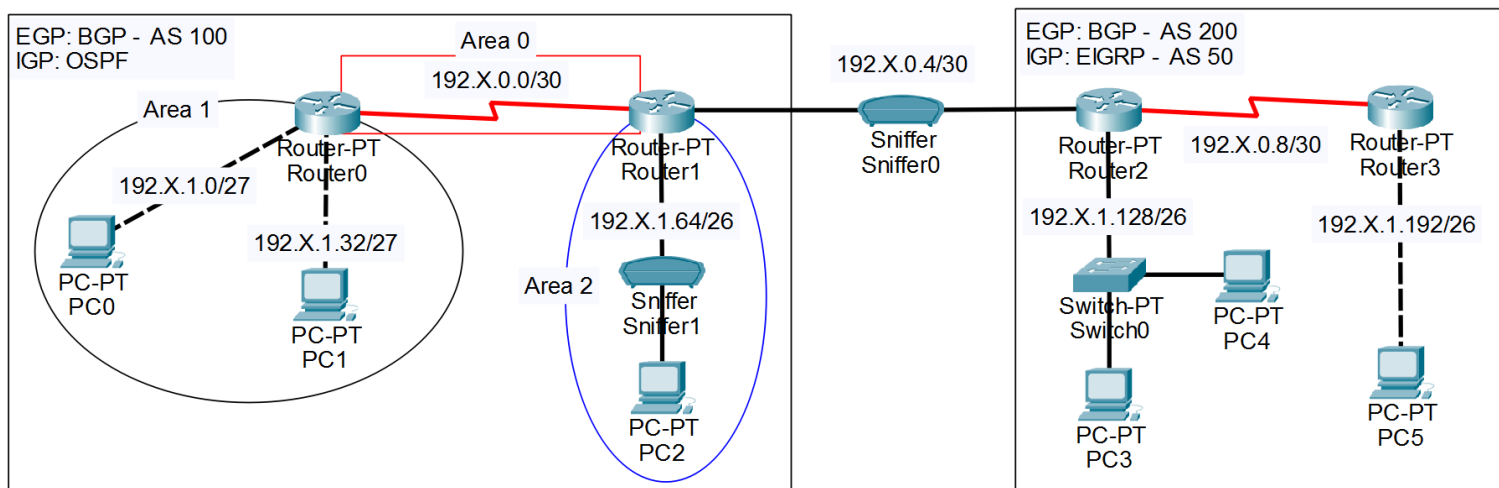


Figure 4: Dynamic routing topology.

Assign appropriate IP addresses to all PCs and routers, ensuring that each device receives a unique address within the valid host range of its respective network. The value of **X** in the network address corresponds to the *last two digits* of your student ID. For example, if the network address is **192.X.1.64/26** and your student ID is **1230105**, then **X = 05**, resulting in the network address **192.5.1.64/26**.

Notes:

1. The subnet **192.X.1.64/26** includes IP addresses from *192.X.1.64* to *192.X.1.127*, where *192.X.1.64* is the network ID and *192.X.1.127* is the broadcast address.
2. It is recommended to assign the first available host address (e.g., *192.X.1.65*) to router interfaces as gateways.
3. PCs should use addresses from the remaining usable range (e.g., *192.X.1.66*, *192.X.1.67*, etc.) for proper network configuration.
4. To prevent conflicts, avoid assigning duplicate IP addresses and always verify connectivity after completing the configuration.

5.3.2 Configuring OSPF Routing

We will configure the **OSPF** routing protocol for *AS 100*, as discussed in *EXP. No. 4, Section 4.3.3*. For example, to configure OSPF on **Router0** with process ID **1**, use:

```
Router(config)# router ospf 1
Router(config-router)# network 192.X.0.0 0.0.0.3 area 0
Router(config-router)# network 192.X.1.0 0.0.0.31 area 1
Router(config-router)# network 192.X.1.32 0.0.0.31 area 1
```

Now, configure OSPF on **Router1** within AS 100. **Note:** Do **not** include the *BGP link* (i.e., network 192.X.0.4/30) in Router1's OSPF configuration.

5.3.3 Configuring EIGRP Routing

We will configure the **EIGRP** routing protocol for *AS 200*, as discussed in *EXP. No. 3, Section 3.3.2*. For example, to configure EIGRP on **Router3** with EIGRP AS number **50**, use the following commands:

```
Router(config)# router eigrp 50
Router(config-router)# network 192.X.1.192 0.0.0.63
Router(config-router)# network 192.X.0.8 0.0.0.3
Router(config-router)# no auto-summary
```

The command `no auto-summary` is used to disable EIGRP automatic summarization, ensuring that subnetted routes are propagated correctly.

Next, configure EIGRP on **Router2** within AS 200. **Note:** Do **not** include the *BGP link* (i.e., network 192.X.0.4/30) in Router2's EIGRP configuration.

5.3.4 Configuring BGP Routing

BGP configuration is required only on **Router1** and **Router2**. The first step is to enable the **BGP process** and specify the router's AS number. For example, to enable **BGP** on *Router1* with AS 100, use the following command:

```
Router(config)# router bgp 100
```

At this point, BGP is enabled on *Router1*. The next step is to establish a neighbor relationship with the router in the other AS (an eBGP peer). For instance, to configure *Router1* to peer with a neighbor at IP address **192.X.0.6** in AS 200, use:

```
Router(config-router)# neighbor 192.X.0.6 remote-as 200
```

Repeat the same steps on **Router2** with the appropriate IP address and AS numbers.

5.3.5 Redistributing BGP into OSPF

To allow OSPF to learn routes from BGP, use the `redistribute` command. For example, on **Router1** with OSPF process ID **1**, use the following configuration:

```
Router(config)# router ospf 1
Router(config-router)# redistribute bgp 100 subnets
```

5.3.6 Redistributing OSPF into BGP

To allow BGP to learn routes from OSPF, also use the `redistribute` command. For example, on **Router1** with OSPF process ID **1**, configure as follows:

```
Router(config)# router bgp 100
Router(config-router)# redistribute ospf 1
```

5.3.7 Redistributing BGP into EIGRP

When redistributing BGP routes into EIGRP, you must specify metric values that correspond to the interface through which the routes enter the EIGRP domain. In this topology, **Router2** connects to **Router3** via a **Serial link**, which is used to inject the redistributed BGP routes. Therefore, the *metric values of the Serial interface* should be applied during redistribution.

For example, on **Router2** (EIGRP AS **50**), the Serial link may have the following characteristics (verify using `show interfaces` before setting metrics): (i) Bandwidth = 1544 Kbps, (ii) Delay = 20000 μ s (use 2000 in the command, as delay is entered in tens of microseconds), (iii) Reliability = 255/255, (iv) Load = 1/255, and (v) MTU = 1500 bytes. Use the following commands:

```
Router(config)# router eigrp 50
Router(config-router)# redistribute bgp 200 metric
                        1544 2000 255 1 1500
```

5.3.8 Redistributing EIGRP into BGP

To allow BGP to learn routes from EIGRP, use the `redistribute` command. For example, on **Router2** with EIGRP AS number **50**, enter these commands:

```
Router(config)# router bgp 200
Router(config-router)# redistribute eigrp 50
```

5.3.9 Configuring BGP Timers

To set the BGP Keepalive and Hold-time timers for **Router1** to 30 and 90, respectively, use the following commands:

```
Router(config)# router bgp 100
Router(config-router)# timers bgp 30 90
```

The aforementioned command will change the timers on the router. If the configured hold timers between two peers differ, the peer session will still be established, and the *smaller timer* value will be used.

5.3.10 Verifying BGP Routing and Neighbor Information

Once **BGP** is configured, it is essential to verify the routing setup and peer connections to ensure proper network communication. The following commands help inspect the BGP configuration and monitor its behavior:

- Display the routing table:

```
Router# show ip route
```

This command displays the router's entire routing table, including *BGP-learned routes*, *OSPF*, *EIGRP*, and directly connected networks. It also provides information on network prefixes, administrative distances, and next-hop addresses, helping verify if BGP routes are installed correctly. Figure 5 shows the output of this command for *Router1*, which is an *Autonomous System Boundary Router (ASBR)*.

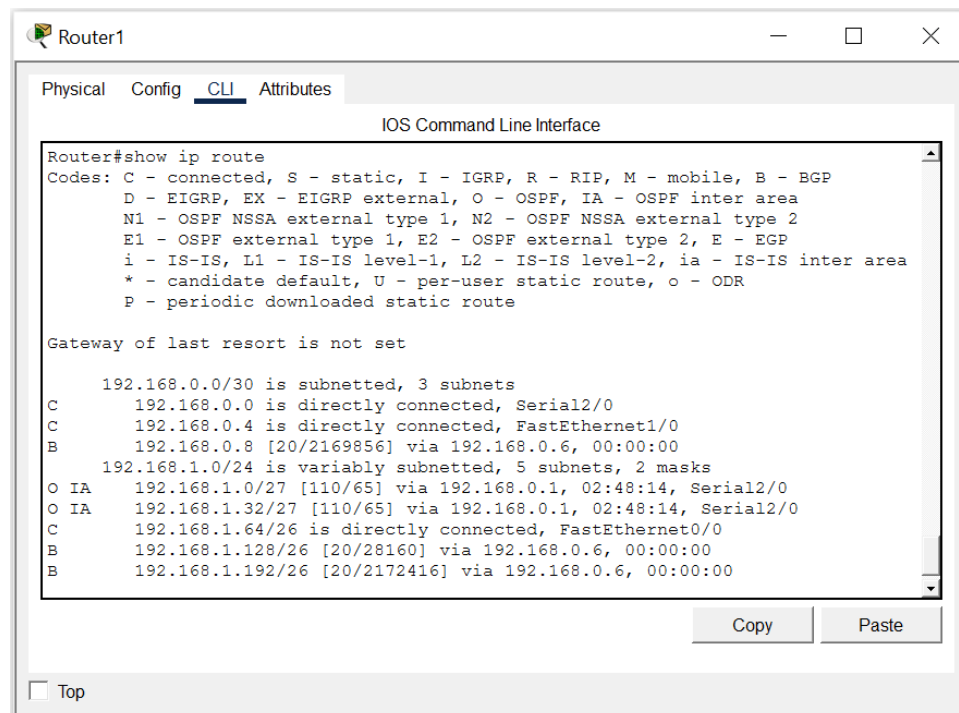


Figure 5: Routing table of Router1, an ASBR (Inter-AS router).

In the routing table, entries marked with “**B**” indicate routes learned via **BGP**. This means that the router is exchanging routing information with networks in

other ASes through BGP. Routes within the same AS are learned using **OSPF** and are marked with “**O**” in the table. Additionally, the administrative distance is displayed for each route:

- **BGP-learned routes** have an administrative distance of **20**, making them highly preferred for external routing.
- **OSPF-learned routes** have an administrative distance of **110**, indicating they are used for internal routing within the AS.

Figure 6 shows the routing table output for **Router0**, which is an internal router. We can notice that routes learned through **OSPF** are marked with “**O**”, indicating that the router is using OSPF for internal routing within the AS. Additionally, some routes are labeled with “**O E2**”, which means these networks belong to a different AS and were learned as external routes via **OSPF Type 2 (E2)**. These external OSPF routes are redistributed into OSPF by the *ASBR*. This routing table confirms that Router0 relies entirely on **OSPF** for internal communication while still learning about external networks through **OSPF Type 2** routes redistributed from **BGP** by the *ASBR*.

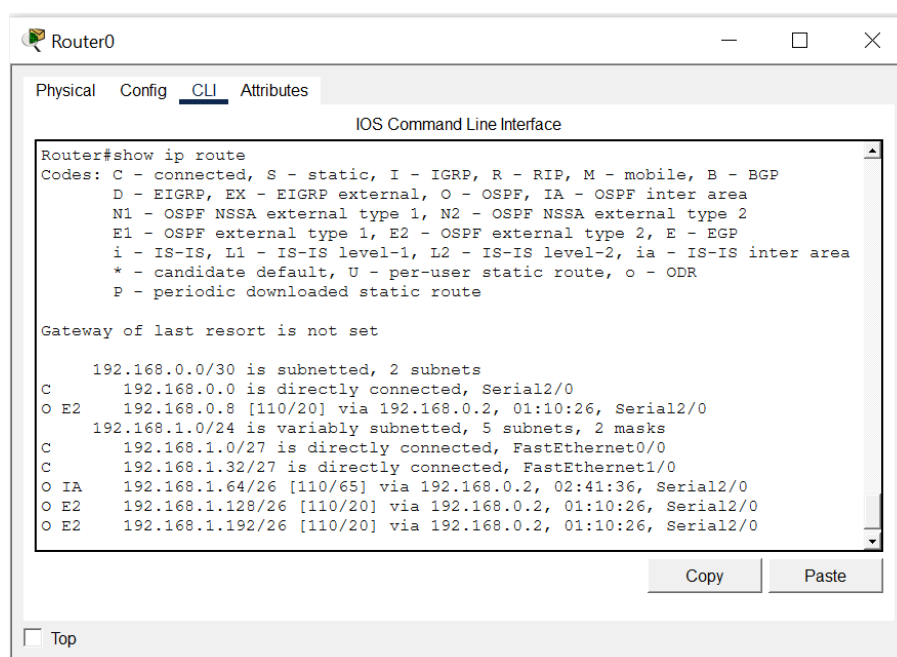


Figure 6: Routing table of Router0, an internal router (Intra-AS router).

Figure 7 shows the routing table output for **Router3**, which is part of *AS 200* and runs **EIGRP**. Routes marked with “**D**” are internal EIGRP routes, indicating networks that belong to the same AS and are learned directly through EIGRP. In contrast, routes marked with “**D EX**” represent external EIGRP routes, which are redistributed from another routing process—in this case, BGP—into EIGRP. The administrative distance values confirm this distinction: internal routes have an AD of **90**, while external routes carry a higher AD of **170**, making them less preferred. This routing table demonstrates that Router3 relies primarily on EIGRP for internal communication within AS 200, while still learning about external networks through redistributed BGP routes marked as EIGRP external.

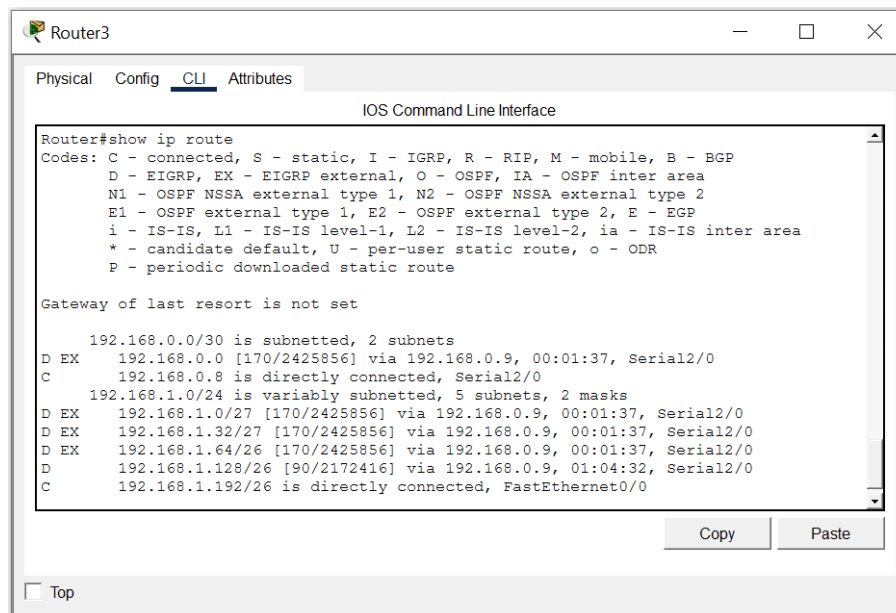


Figure 7: Routing table of Router3, an internal router (Intra-AS router).

- Show BGP neighbor relationships and summary:

```
Router# show ip bgp summary
```

This command provides a summary of all BGP neighbors, displaying details such as the peer AS number, peer state, number of prefixes received, and session uptime. It is useful for troubleshooting BGP connections and ensuring proper neighbor relationships. Figure 8 shows the BGP summary table output for Router1.

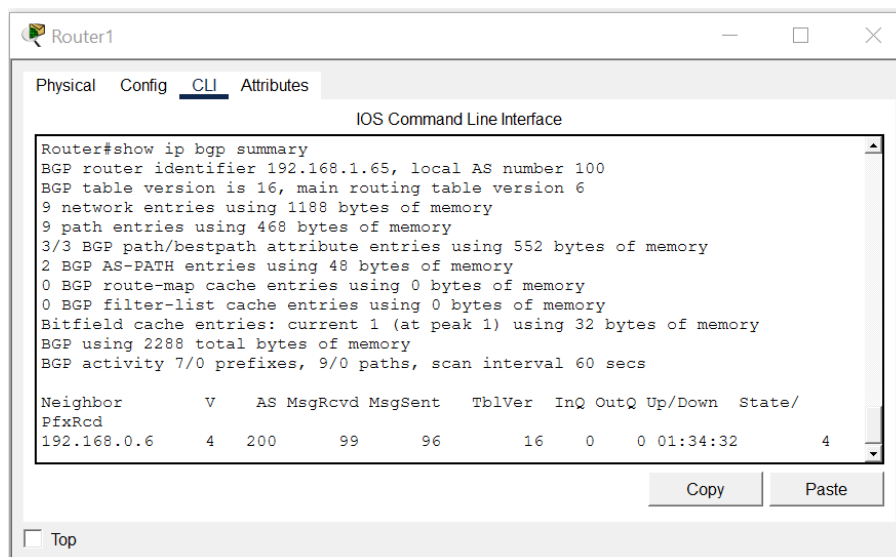


Figure 8: BGP neighbor summary for Router1.

- Show detailed BGP neighbor information:

```
Router# show ip bgp neighbors
```

This command provides in-depth details about each BGP neighbor, including the remote AS number, BGP version, session state, and timers. It is particularly useful for diagnosing BGP neighbor relationships and ensuring stable peer connections.

Figure 9 presents an example output from this command on Router1. The key details in the output include:

- **BGP Neighbor State:** Indicates that the BGP session is in the *Established* state, signifying a successful connection.
- **Remote AS Number:** Displays the AS number of the BGP neighbor, which is 200. This helps verify peering relationships.
- **Remote Router ID:** Shows the unique identifier of the BGP neighbor, which is 192.168.1.129. The Router ID is typically an IP address used to distinguish the BGP router in the network.
- **Received and Advertised Prefixes:** Shows the number of routes received (3) and sent (5) to the BGP neighbor.
- **Message Statistics:** Displays BGP message exchange details, with 30 messages sent and 39 received. These messages include Open, Notification, Update, and Keepalive messages, which are essential for maintaining BGP session stability and exchanging routing information.

```

Router1
Physical Config CLI Attributes
IOS Command Line Interface

Router#sh ip bgp neighbors
BGP neighbor is 192.168.0.6, remote AS 200, external link
BGP version 4, remote router ID 192.168.1.129
BGP state = Established, up for 00:17:13
Last read 00:17:13, last write 00:17:13, hold time is 180, keepalive interval is 60 seconds
Neighbor capabilities:
  Route refresh: advertised and received(new)
  Address family IPv4 Unicast: advertised and received
Message statistics:
  InQ depth is 0
  OutQ depth is 0

      Sent      Rcvd
Opens:          1          1
Notifications:  0          0
Updates:        11         18
Keepalives:     18         19
Route Refresh:  0          1
Total:          30         39
Default minimum time between advertisements runs is 30 seconds

For address family: IPv4 Unicast
BGP table version 23, neighbor version 6/0
Output queue size : 0
Index 1, Offset 0, Mask 0x2
1 update-group member

Prefix activity:
Prefixes Current:      5          3 (Consumes 184 bytes)
Prefixes total:        5          3
Implicit Withdraw:      0          0
Explicit Withdraw:      0          0
Used as bestpath:      n/a        1
Used as multipath:      n/a        0

      Outbound   Inbound
Local Policy Denied Prefixes:  -----
Total:                        0          0
Number of NLRI's in the update sent: max 3, min 1

Address tracking is enabled, the RIB does have a route to 192.168.0.6
--More--

```

Figure 9: Detailed BGP neighbor information for Router1.

5.3.11 Packet Sniffing

To analyze network traffic and understand data flow, we use the Sniffer tool to capture and inspect packets. In this setup, **Sniffer0** is used to capture traffic between routers from different ASes, while **Sniffer1** captures traffic within the same AS. This allows us to observe both inter-AS and intra-AS communications.

To enable packet capture, follow these steps:

- Navigate to Sniffer → GUI → Service.
- Toggle Check On/Off to enable or disable packet capturing.
- Select Port0 or Port1 to choose which interface captures incoming packets, allowing you to control the direction and source of traffic.

By default, the sniffer captures all protocols, but students can refine their analysis by applying filters. To clear all captured packets, simply click the “Clear” button, which removes the existing packet data from the capture window. For a more focused inspection of network traffic, students can apply specific filters by selecting “Edit Filters” and choosing the desired protocols or packet types to monitor. For this experiment, we focus on the following protocols:

- **ICMP**: Used for ping to test connectivity between devices.
- **OSPF** and **BGP**: Routing protocols that display how network routes are exchanged and updated.

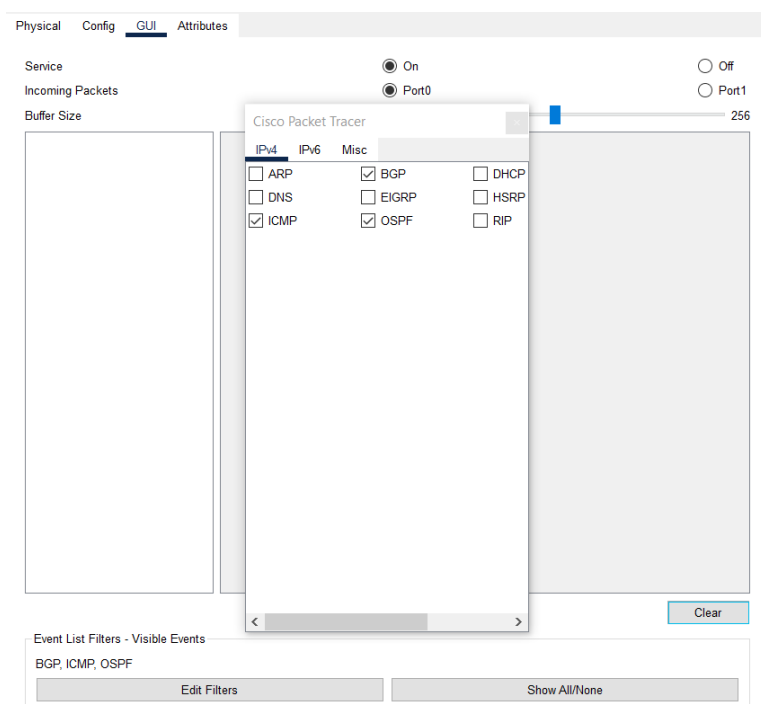


Figure 10: Applying filter to capture specific protocols.

Figure 10 demonstrates how to apply protocol filtering. After setting the filters, send a packet from **PC2** to **PC3** (located in different ASes) and inspect it using the sniffer tool. Clicking on the captured packet displays key details, including: (i) source MAC address, (ii) destination MAC address, (iii) source IP address, and (iv) destination IP address.

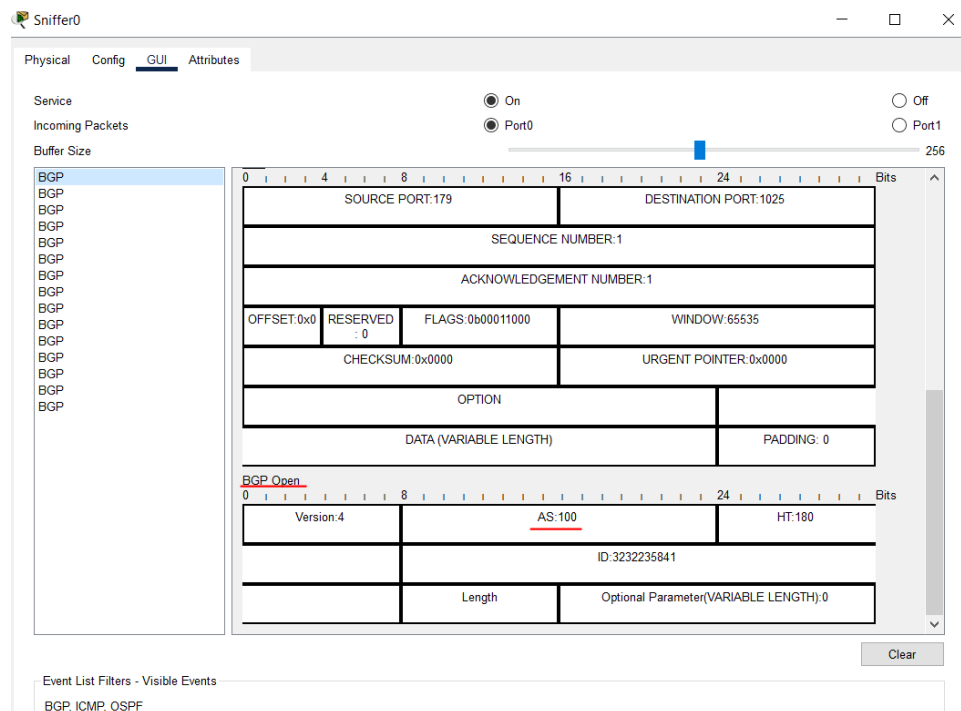


Figure 11: BGP OPEN message.

Sniffer0, positioned between routers belonging to different ASes, is responsible for capturing BGP traffic. This is because BGP is used to exchange routing information between ASes, and the sniffer will capture the BGP updates. Figure 11 illustrates the structure of a BGP OPEN message, which is the initial step in establishing a BGP session between neighboring routers. The OPEN message includes information that must be negotiated and mutually accepted by both BGP routers before routing information can be exchanged. This information includes:

- BGP Version Number (Version:4): Indicates the BGP version used by the router. Version 4 is the most widely deployed and current version. A version mismatch between peers prevents session establishment.
- AS Number (AS:100): Identifies the router's autonomous system. This value also determines whether the BGP session is internal (iBGP) or external (eBGP).
- Hold Time (HT:180): A timer that ensures the BGP neighbor is still responsive. Cisco routers, by default, use a hold time of 180 seconds.
- BGP Identifier (ID:3232235841): A unique 32-bit value identifies the BGP router in advertised prefixes. It also helps prevent routing loops within an AS. The BGP ID must not be zero and can be manually set using the `bgp router-id` command.
- Optional Parameters: Define the session capabilities of the BGP router. They allow new features to be added without requiring a change in the BGP version number.

Figure 12 shows the details of a BGP UPDATE message, which are used for advertising and exchanging routing information between BGP neighbors. The UPDATE message includes only the prefix in the message. UPDATE messages also act as keepalives to lessen unnecessary traffic.

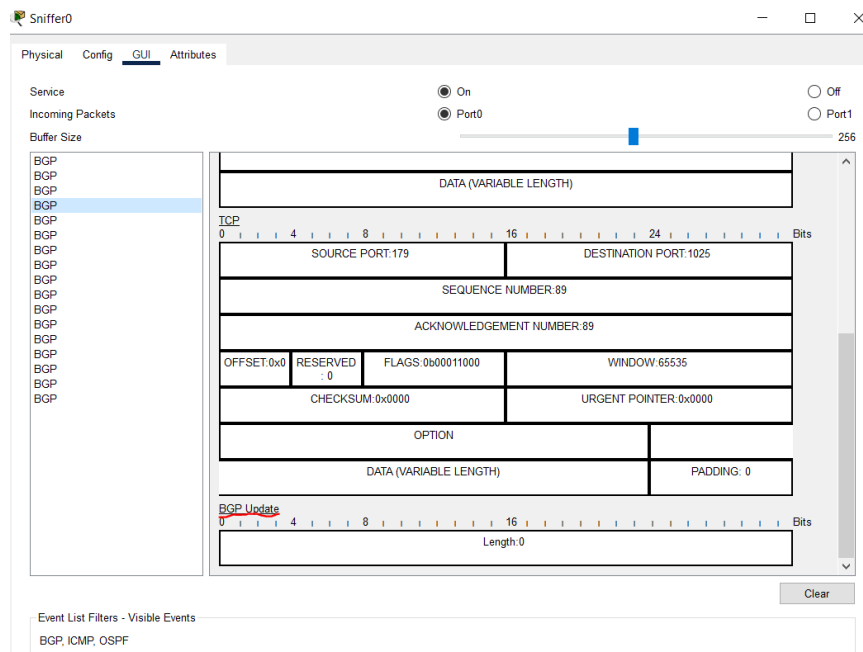


Figure 12: BGP UPDATE message.

In Cisco Packet Tracer's simulation mode, the *Keepalive* message can be captured at Layer 5 (the Session Layer) of the OSI model. This is because BGP operates over TCP, which uses port 179 at Layer 4 (the Transport Layer). After the initial TCP three-way handshake is completed, Keepalive messages are exchanged to ensure that BGP neighbors remain active. These messages are lightweight, typically consisting of only a header without any payload, and are sent at regular intervals—by default, every 60 seconds—to verify that the peer is still responsive. If a router fails to receive a response after three consecutive Keepalive messages, the peer is considered inactive. This duration ($3 \times 60 = 180$ seconds) is known as the Hold Time. If no Keepalive or other BGP message is received within this period, the BGP session is terminated. Figure 13 shows the structure of a Keepalive message captured during the simulation.

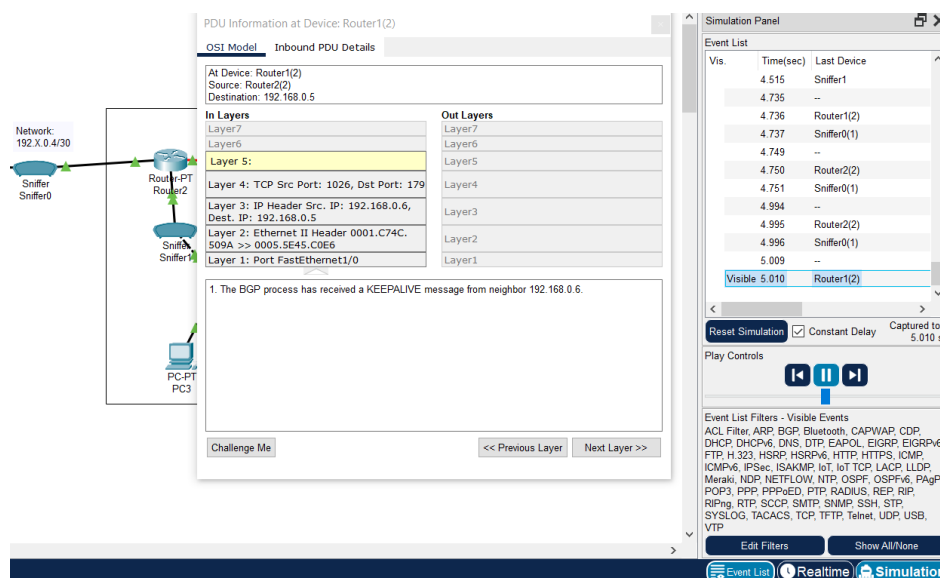


Figure 13: BGP Keepalive message.

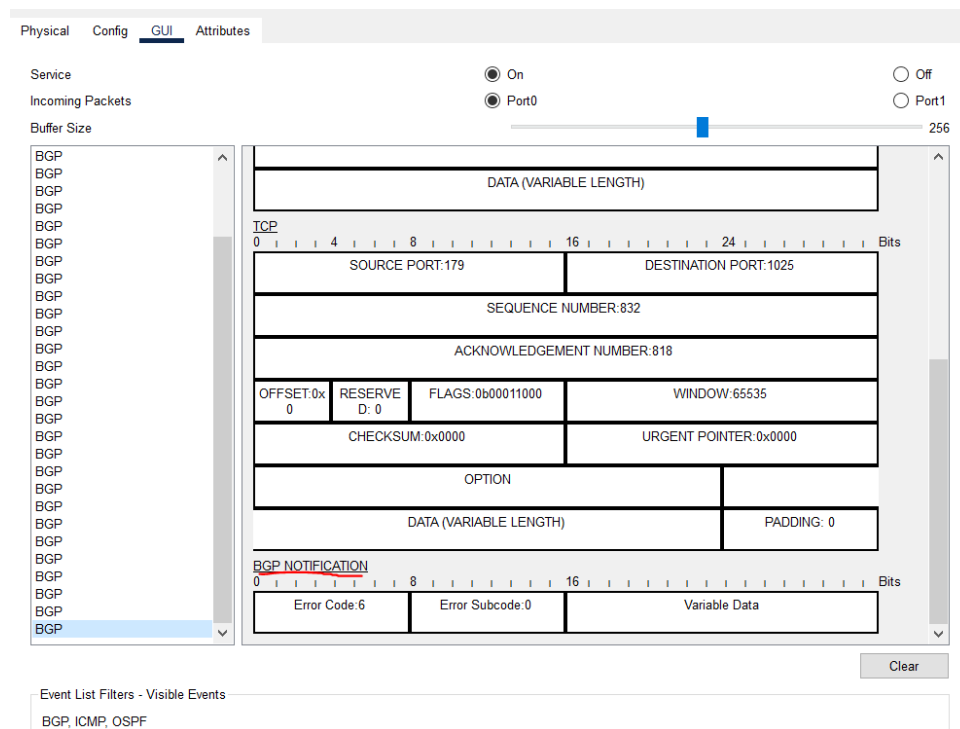


Figure 14: BGP Notification message.

Figure 14 displays the structure of a BGP *Notification* message, which is sent when an error occurs or when a problem arises during the BGP session. This message informs the peer of issues such as misconfigurations, protocol violations, or connection failures. It contains an error code, an error subcode, and, in some cases, additional data providing more context about the issue. When a router needs to terminate the session due to an error, it sends a Notification message to help the peer identify the cause of the problem.

Sniffer1, located within AS 100, will capture **OSPF** and **ICMP** traffic. OSPF packets are exchanged within an AS to update the routing table. In this case, the source of the OSPF packet is the default gateway of the network, and the destination is a multicast address (224.0.0.5), as shown in Figure 15.

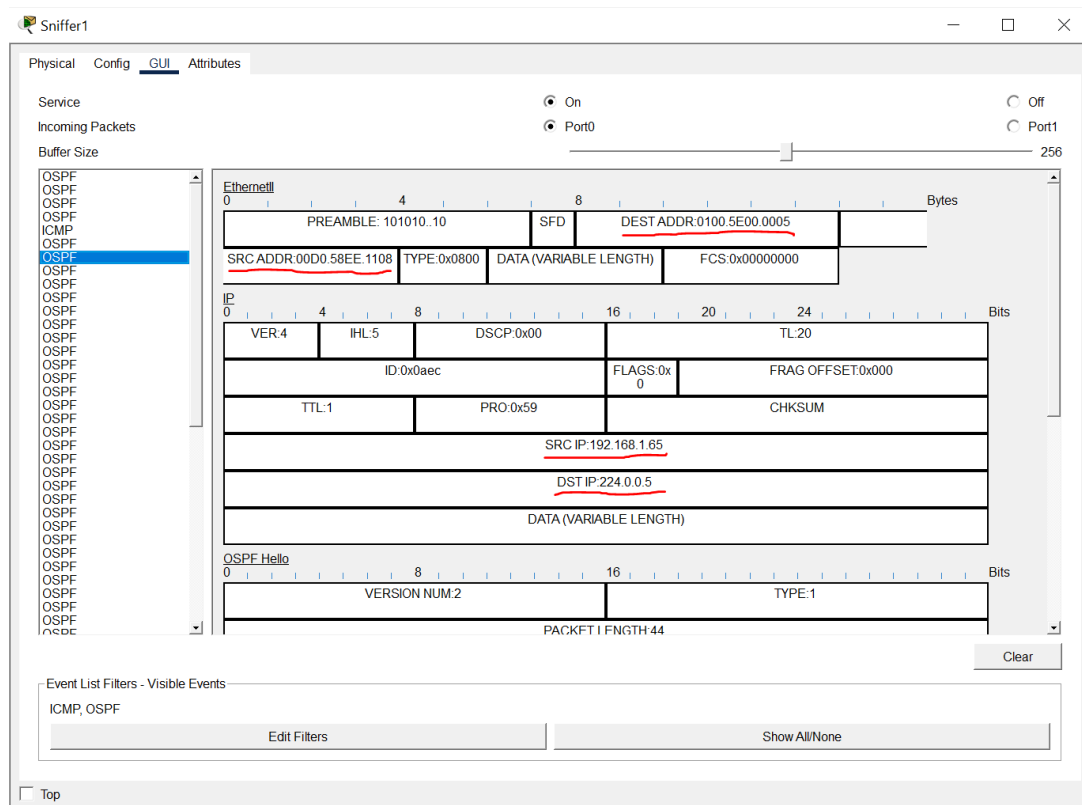


Figure 15: Packet details for OSPF protocol.

5.4 ToDo

This section will be provided by the instructor.